Market research of the space industry:

Introduction:

The Apollo mission, which sent mankind on the moon for the first time, cost 275 billion US dollars (adjusted for inflation)[1]. This is more than most countries' annual budget. Space exploration is probably one of the industries with the highest barriers to entry, due to the cost of even the simplest mission. Thus, cooperation to finance and organize such operations typically involve numerous governmental and private actors, from multiple countries. The Apollo mission's main investor was the US government, who had a very clear incentive to help this mission succeed: win the race to space, one of the many aspects of the Cold War. With this successful mission, the USA significantly outpaced the USSR in space exploration. Space exploration has an essential political aspect, with the unfathomable sums of money required and the almost ineluctable government participation. The International Space Station (ISS) is a more recent example of this fact. The ISS is funded by government agencies, mostly NASA (US) and the ESA (EU), for a total of sixteen countries. At any time, multiple nationalities are present in the ISS, and any diplomatic conflict between some of the nations responsible for the ISS may lead to delicate situations. The decision to include certain countries or not in the ISS project is clearly partially political, such as the decision to not let China partake in any program managed by NASA, which led to China creating their own space station, and by extension, their own space strategy.

The space expansion consumes an extraordinary amount of hardware, which adds another international cooperation layer: a significant portion of this hardware is semiconductors, that require rare-earth elements (REEs). Few countries extract REEs and few are those that also manufacture semiconductors.

There are currently multiple actors in the space expansion world, the biggest ones by far being the US+EU group and China. However, their strategies for progress and obtaining essential components are radically different.

Part I: The current state of the european space strategy

1. Europe as space power

Context: structure and main actors

As of today, the European Union (EU) is a major global space power, operating critical satellite infrastructures. It is undeniable that space is increasingly recognised by the EU as a strategic domain for security, defence as well as technological sovereignty. In fact, space infrastructure is essential not only for communications (e.g. IRIS² satellite constellation that

[1] https://www.planetary.org/space-policy/cost-of-apollo

provides secure communication infrastructure for governments and defence actors), navigation (e.g. The Galileo navigation system provides Europe with an independent satellite positioning capability), defence capabilities (e.g. the program GovSatCom for security operations) and economic competitiveness (e.g. the Copernicus Earth observation programme that provides earth observation data supporting multiple industries).

In order to ensure structure within the space sector, multiple governance layers exist. Firstly, European Union institutions set the overall policy framework and funding programmes for space activities while integrating space policy with broader economic, security, and digital strategies. Moreover, the European Space Agency (ESA) coordinates technical development, research, and mission implementation. Lastly, national space agencies implement national space policies and contribute to funding European programmes.

The space policy is increasingly linked to EU strategic autonomy, reducing dependence on non-European infrastructure. In this context, the EU space strategy for security and defence can be introduced.

EU space strategy for security and defence (2023)

The EU Space Strategy for Security and Defence (2023) aims not only to protect EU space assets by putting into place infrastructure that ensures tracking and monitoring of objects in orbit, but also to deter hostile activities in space by developing secure and resilient satellite systems. It overall contributes to strengthening Europe's strategic autonomy by maintaining independent access to space through European launch capabilities hence enabling Europe to launch its own satellites without relying on foreign providers.

The strategy reflects growing geopolitical competition in space and the increasing militarisation of the domain.

Hence, the EU space strategy implements several major policy pillars. First, it seeks to enhance the resilience and protection of space systems by strengthening safeguards against cyberattacks, signal jamming, and potential anti-satellite threats targeting European orbital infrastructure. Second, the strategy emphasises the development of space domain awareness, improving Europe's capacity to monitor orbital activity, track space debris, and detect potential risks to satellites operating in increasingly congested orbital environments. In addition, the EU promotes the use of space for security and defence purposes by expanding technologies capable of supporting both civilian applications and military operations. Another important pillar is related to partnerships and international cooperation, particularly through closer coordination with NATO and other allied actors in order to address shared security challenges in space. Finally, the strategy supports industrial and technological development by encouraging innovation, fostering space sector startups, and strengthening research collaboration between the defence and space industries to ensure Europe remains competitive in the evolving global space economy.

2. The economic importance of the european space sector

The growth of the European space economy is more and more relevant: the development of satellite communications, earth observation data services as well as navigation systems is the main source of this growth.

In fact, space infrastructure supports key economic sectors such as transportation as they enable precise positioning and timing services used in aviation, maritime navigation, and road transport thus improving safety. But also agriculture as earth observation satellites provide high-resolution data on soil conditions, crop health, and weather patterns, allowing farmers to implement precision agriculture techniques that optimize yields and reduce resource consumption. Moreover, climate monitoring has improved due to space infrastructure since satellite based observation systems collect environmental data on atmospheric composition, ocean temperatures, and ice coverage, supporting scientific research and informing policies aimed at addressing climate change. Lastly, as explained previously, defence systems benefit a lot from space infrastructure.

Overall, the EU strategy emphasizes industrial competitiveness and the development of a strong European supply chain to compete with American private sector industries such as SpaceX or the overall NASA ecosystem and China's state driven space program (c.f. Section 2).

3. France and Germany's space strategies

France's space strategy

Within the EU space programs, France is one of the leading contributors to the European space sector and plays a central role in ESA. It aims to ensure autonomous and sustainable access to space for Europe by prioritizing technological sovereignty, industrial competitiveness as well as security and resilience of space infrastructure. For instance, France hosts major companies such as Airbus Defence and Space, Thales Alenia Space or Arianespace. Moreover, the french space agency (CNES) is one of the most influential ones in Europe, helping France reach its strategic approach that favors strong European integration, strategic autonomy from non-European technologies and increased EU level industrial policy. Lastly, besides its will to contribute to Europe's space sovereignty and strengthening its infrastructures by making them as independent as possible, it also aims to develop active defence measures in space to protect satellites and to increase funding and investment in national and European space programs.

Germany's space strategy

On the other hand, Germany focuses strongly on industrial innovation and commercial space applications. It prioritises the strengthening of Germany's position as a leading space

technology and research nation, the promotion of commercial space innovation and private sector growth and the support of European cooperation in space programs.

In fact, German government funding for civil space programmes has increased significantly since 2010. Their main focus is satellite technologies, climate monitoring and the development of earth observation systems. The German Space Safety and Security Strategy (2025) focuses on improving space defence capabilities and ensuring operational capability in space during crises and conflicts. It seems to be cooperation with NATO and EU partners in space security but less integrated than France's strategy. Its main institutions include DLR (German Aerospace Center) as well as the Federal Ministry for Economic Affairs and Climate Action that is highly involved within the Safety and Security Strategy programs.

France vs Germany: what are the key differences?

Both countries' strategies differ on many points.

Strategically, France emphasizes strategic autonomy and defence capabilities and supports stronger EU-level space integration whereas Germany focuses more on industrial competitiveness and commercial space development.

In terms of policy, France has a relatively important state involvement whereas Germany focuses on public-private partnerships and innovation ecosystems.

When it comes to security, France is more proactive in developing space defence capabilities whereas Germany seems to be more cautious and slow.
Lastly, when it comes to their role within the EU, both countries are the largest contributors to ESA and play central roles in EU space initiatives. However, France pushes strategic independence from external actors whereas Germany prioritizes economic competitiveness and technological innovation.

Part II: The Chinese space strategy

While the Western space strategies discussed in Section 1 rely on international partnerships and shared commercial contractors, China's spatial enterprise is defined by a national approach of missions.

A small history of the Chinese space program

China started space exploration in the 1970s, with its first satellite, Dong Fang Hong I, launched in 1970, making China a space pioneer . This was part of the "Two Bombs, One Satellite" plan, which also led to the making of Chinese nuclear weapons. In the 1990s,

human space exploration was discussed with the Shenzhou project (also called Project 921). In October 2003, Yang Liwei, the first taikonaut, was launched into space during the Shenzhou 5 mission.

The project of a Chinese space station was concretized after China was denied access to the ISS or any other NASA project by the USA through the Wolf Amendment in 2011. Two labs were then launched in the 2010s, Tiangong-I and Tiangong-II, to prepare for the final spatial station, Tiangong ("Heavenly Palace"). Tiangong is composed of multiple modules: Tianhe, the core module, launched in 2021, and Wengtian and Mengtian, the lab modules, in 2022. Tiangong, being an exclusively Chinese project, was built without international cooperation. Although it would not benefit immediately from potential foreign technological progress, it gives the Chinese space agency total freedom on the research it wants to conduct, without any friction from partners.

Semiconductor supply chain:

All of the projects mentioned above rely heavily on semiconductors: the quantity and quality of that component defines the probability of success of any space mission. However, the supply of that component is a crucial matter: it depends on the supply of REEs.

Space projects are thus particularly vulnerable to the price, availability and technological properties of semiconductors. If their price increases, the whole mission may become too expensive. If they become unavailable (typically due to REEs scarcity), the mission is simply undoable. About technological properties, semiconductors for space must be "radiation-hardened" (rad-hard) in order to survive long enough to the exposition of space radiation.

Past Chinese semiconductor supply chain (1965-2020s):

The first Chinese integrated circuits were produced in the 1960s, during the Twelve-Year Science plan, only a few years after the first integrated circuit was developed in the US in 1958. Semiconductor development was an essential part of the "Two Bombs, One Satellite" plan, where this component was crucial to a smooth running of the space operations. However, it stayed at the research level for more than one additional decade, not being produced on a large scale.

In 1996, the first stable semiconductor industrial producer, Hua Hong Semiconductor, was created. SMIC (Semiconductor Manufacturing International Corporation), the current leader, was started in 2000. Some companies started producing semiconductors, for example HiSilicon (from 2004), a branch of Huawei. Simultaneously, various foreign investors, such as Intel or Samsung, relocate their semiconductor factories in China.

At this point, China had launched multiple space missions, relying on both locally-produced chips and foreign chips. Overall, the Chinese semiconductor imports were increasing despite the expansion of local producers.

In 2014, due to worries about the increasing imports of semiconductors, the China Integrated Circuit Industry Investment Fund was created, with the goal of reducing foreign imports of chips. The funds were dedicated to buy equipment and materials to strengthen the supply chain.

Current supply chain (2020s):

A crucial point to become less dependent is the sourcing of machines to produce the semiconductors. The main producer of the lithography machines, by far, is the Dutch company ASML (Advanced Semiconductor Materials Lithography), an actual monopolist for the most advanced models.

To solve this supply issue, China is relying on both innovation and mandatory national production.
Since 2025, the Chinese government requires companies to reach a threshold of 50% of domestically sourced materials for any new equipment, in order to increase supply chain resilience. The goal is to become less dependent of foreign chip companies, leading to the rise of companies such as SMIC (Semiconductor Manufacturing International Corporation) or Naura technology. Their new goal is to stabilize the production of smaller chips, which are overall more efficient.

When it comes to recent innovation, in early 2026, researchers of Fudan university claimed to have developed radiation-resistant semiconductors that allow communication systems that can tolerate the space environment better, with a life expectancy of 271 years. This would solve one of the main challenges of space research: the life expectancy of components being lesser than the duration of a mission.

Part III: What's next for Europe

1. Strengthening Europe's strategic and economic position in the space sector

Strengthening European strategic autonomy

According to the European Parliament Research Service in 2023, Europe must reduce dependence on foreign space technologies and launch services, particularly from the United States and emerging competitors such as China. As strategic autonomy requires greater coordination between EU institutions, ESA, and national space agencies(to avoid fragmentation of resources) , the EU should strengthen its independent access to space, ensuring reliable European launch systems and infrastructure.

Hence, the EU must invest in critical technologies and components to avoid reliance on non-European supply chains in order to ensure resilience within its space infrastructure, protecting satellites from cyber threats and geopolitical disruptions.

Expanding the European space economy

ESA Space Economy as well as the OECD emphasize that the global space economy is expected to grow significantly in the coming decades, hence creating opportunities for European industry: in fact, space infrastructure will increasingly support sectors (mentioned previously in section 1)such as telecommunications, climate monitoring, transportation, etc.

It is implied that future economic growth will be driven by "downstream services", specifically data analytics and digital applications derived from satellite systems. In order to achieve that, public investment will be a key element in supporting the development of these new "space markets": these investments should target high value space services.

Supporting private investment and the "new space" economy

To be more specific about public investments, the European Investment Bank affirms that Europe must expand financial instruments to support space startups and private space companies as access to venture capital remains significantly lower in Europe than in the United States for example, hence limiting the growth of innovative space companies.

Consequently, Europe should encourage public/private partnerships to accelerate innovation in launch services, satellites, and data applications since the development of a strong startup ecosystem is necessary to remain competitive in the global space industry.

2. Governance, sustainability and international cooperation in Europe's future space policy

Strengthening space innovation

OECD reports published in 2021 found that space technologies generate significant spillover benefits for other industries, including telecommunications, materials science, and robotics. Consequently, many experts think that governments should promote technology transfer from space programs to commercial markets and that innovation policies should support collaboration between research institutions, universities and private companies and that Europe should expand programs that encourage commercialisation of space technologies developed in publicly funded missions.

Improving governance and coordination in the European space sector

Nevertheless, the European space sector remains institutionally fragmented, with overlapping roles between ESA, the EU, and national agencies. In its report entitled *ESA strategy and the future of the European space sector* published in 2024, the Robert Schuman Foundation emphasizes that future space policy should focus on stronger institutional coordination to maximize efficiency and avoid duplication.

For now, ESA's long-term strategy emphasizes sustainability of space operations, resilience of space infrastructure and an attempt on technological leadership but long term strategic priorities are still to be defined in order to ensure actual space leadership while encouraging sustainable space practices. In fact, an increasing number of satellites and debris pose growing risks to space operations making the investment in technology for space traffic management another key element for this new space.

International cooperation

Lastly, the OECD as well as the ESA and the European Parliament Research Service emphasize that, despite pursuing strategic autonomy, Europe should maintain strong international partnerships.
In fact, cooperation with global partners supports scientific missions, planetary exploration and technological innovation. This could enable Europe to position itself as a leader in global space governance, particularly in areas such as sustainability and responsible space operations.

Conclusion:

In the 2020s, space expansion has moved beyond the Cold War's "race to space" toward a complex competition of habitation, hardware and infrastructure.There are many more contestants, and as missions become more complex, international alliances are inevitable for most. The primary strategy has shifted from pure rocket propulsion to the durability of the supply chain of essential materials and the progress in their characteristics.
Two main divergent philosophies have emerged. The European strategy (or Western overall) relies on distributed innovation, with numerous governmental actors and private participants in a competitive market. This collaborative approach allows for a wider range of intellectual and material resources, but could lead to decisional issues during missions or even diplomatic conflicts. On the other hand, China has pursued an opposite approach: developing an independent space program, from the planning of the missions, to the research priorities and the hardware supply chain. Europe must now decide either to remain the main ally of the increasingly volatile US or to develop its own strategy by securing their own supply chain, thus becoming a third pole in the current bipolar space industry.

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